

Compositions of Transformations and Symmetry

Answer Key

High School Geometry

Quick Answers

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|--|---|
| 1. horizontal component = -5 , vertical component = -3 | 6. 15 |
| 2. 36 | 7. 70 |
| 3. (a) x-coordinate of the final image = -4 , (a) y-coordinate of the final image = 6, — | 8. (a) y-coordinate of the final image = -5 , — |
| 4. 8 | 9. (a) smallest rotation angle = 60 , (b) angle of two turns in succession = 120 , — |
| 5. x-coordinate of the final image = -3 , y-coordinate of the final image = -6 | 10. (a) x-coordinate of the final image = 5, (a) y-coordinate of the final image = -3 , (b) RS = 5, (c) image length = 5, — |
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What is verified

17 of 21 answers machine-checked against SymPy, across 10 problems.
— marks an answer only you can judge — problems 3, 8, 9, 10. The worked solution states what a correct response must contain.

Curriculum

Level: High School Geometry
HSG-CO.A – *problems 1, 2, 3, 4, 5, 6, 7, 8, 9, 10*
Difficulty 1–5 of 5 across 21 tagged checks.

- 1.** Translate by $(x + 3, y - 5)$, then by $(x - 8, y + 2)$. Find the single translation.

Solution: Translations compose by adding their components, because each one shifts every point by a fixed amount and the shifts simply stack.

$$\text{horizontal: } 3 + (-8), \quad \text{vertical: } (-5) + 2$$

So the single rule is $(x, y) \rightarrow (x - 5, y - 3)$: horizontal
and vertical

-5

-3

- 2.** Smallest rotation mapping a regular decagon onto itself.

Solution: A regular n -gon looks identical after a turn of $360^\circ/n$, because its n vertices are equally spaced around a full turn. With $n = 10$: $360^\circ/10$.

36°

3. $P(4, -6)$ reflected in the x -axis then in the y -axis. (a) P'' . (b) The single equivalent motion.

Solution (a): The first reflection negates y , giving $(4, 6)$; the second negates x .

$$P'' = (-4, 6)$$

so the coordinates are

-4

and

6

Solution (b) — instructor-judged. A model response: both coordinates of P have been negated, which is exactly the rule $(x, y) \rightarrow (-x, -y)$ — a half turn, that is a rotation of 180° , about the origin. Reflecting in two perpendicular lines always gives a rotation about their intersection through twice the angle between them, and $2 \times 90^\circ = 180^\circ$.

Full credit: names a rotation of 180° about the origin and cites the negation of both coordinates. *Half credit:* says rotation with no centre or angle. *No credit:* names a reflection or a translation.

4. Reflect in $x = 1$, then in $x = 5$. Find the translation distance.

Solution: Two reflections in parallel lines give a translation perpendicular to them, through twice the gap between the lines — not once.

$$2(5 - 1) = 2(4)$$

So every point moves
units to the right.

8

5. $A(5, -1)$ translated by $(x - 2, y + 7)$, then given a half turn about the origin.

Solution: Do the translation first, then negate both coordinates.

$$A' = (5 - 2, -1 + 7) = (3, 6)$$

$$A'' = (-3, -6)$$

So the coordinates are
and

-3

-6

6. Smallest rotation is 24° . Find the number of sides.

Solution: The smallest rotation is $360^\circ/n$, so read that relation backwards.

$$\frac{360}{n} = 24$$

$$24n = 360$$

$$n = 15$$

So the polygon has

$n = 15$

7. Two reflections in lines through the origin meeting at 35° . Find the rotation angle.

Solution: Reflecting twice in intersecting lines gives a rotation about the crossing point through *twice* the angle between the lines.

$$2 \times 35^\circ$$

So the rotation is

$$\boxed{70^\circ}$$

8. $Q(-2, 5)$ turned a quarter turn counterclockwise about the origin, then reflected in $y = x$.

Solution (a): The quarter turn sends (x, y) to $(-y, x)$, so $Q' = (-5, -2)$. Reflecting in $y = x$ swaps the coordinates, giving $Q'' = (-2, -5)$. The y -coordinate is $\boxed{-5}$

Solution (b) — instructor-judged. A model response: $Q(-2, 5)$ has become $(-2, -5)$ — the first coordinate is unchanged and the second has changed sign, which is the rule $(x, y) \rightarrow (x, -y)$. That is a reflection in the x -axis.

Full credit: names a reflection in the x -axis and cites the unchanged x with the negated y . *Half credit:* says reflection with the wrong mirror line. *No credit:* names a rotation or a translation.

9. Regular hexagon. (a) Smallest rotation. (b) Two of them in succession. (c) Why six return every vertex.

Solution (a): $360^\circ/6$.

$$\boxed{60^\circ}$$

Solution (b): Rotations about the same centre add, so two turns of 60° give $2 \times 60^\circ$.

$$\boxed{120^\circ}$$

Solution (c) — instructor-judged. A model response: six turns of 60° add to $6 \times 60^\circ = 360^\circ$, which is one complete revolution about the centre. A full revolution carries every point of the plane all the way round and back onto itself, so each vertex ends exactly where it began.

Full credit: computes 360° and identifies it as one full revolution returning every vertex. *Half credit:* reaches 360° without the revolution argument. *No credit:* answers only that a hexagon has six vertices.

10. $R(1, 2)$, $S(6, 2)$ reflected in $y = x$, then translated by $(x + 3, y - 4)$; $S''(5, 2)$.

Solution (a): Reflecting $R(1, 2)$ in $y = x$ swaps its coordinates to $(2, 1)$; the translation then adds 3 and subtracts 4.

$$R'' = (2 + 3, 1 - 4)$$

so the coordinates are

and

$$\boxed{5}$$

$$\boxed{-3}$$

Solution (b): R and S share a y -coordinate, so $RS = |6 - 1|$.

$$\boxed{RS = 5}$$

Solution (c): $R''(5, -3)$ and $S''(5, 2)$ share an x -coordinate, so the length is $|2 - (-3)|$.

$$\boxed{R''S'' = 5}$$

Notice the segment turned from horizontal to vertical — the reflection changed its direction but not its size.

Solution (d) — instructor-judged. A model response: a reflection preserves every distance and a translation preserves every distance, so performing one and then the other preserves distance as well. The image of a segment of length 5 therefore has to have length 5; the two answers agreeing is forced, not lucky.

Full credit: identifies both steps as rigid motions and states the distance-preserving property. *Half credit:* says lengths are preserved without identifying both steps. *No credit:* offers the two equal answers as the reason.